

Institute of Actuaries of India

Subject ST6 – Finance and Investment B

September 2017 Examination

INDICATIVE SOLUTION

Solution 1:

i)

- Asian Option (or Average rate option) is an option where pay off at expiry is based on the average stock price during the life of the option, typically computed daily but other rules can apply, in relation to the strike price.
- Barrier Option is an otherwise standard option, which deletes itself if the underlying stock touches a knock-out or activates if it touches a knock-in price.
- Basket Option is an option where underlying "instrument" is a specified basket (or portfolio) of assets.
- Bermuda Option is an otherwise standard option, which can be exercised only on a limited number of dates. It lies in valuation between that of a European Option and an American Option.
- Binary Option is an option with a discontinuous (digital) pay off. The most common form is the "Cash or Nothing" in which the option pays out a fixed amount if the option is in the money at expiry.
- Compound Option is an option on an option. The four main types are call on a call, put on a call, call on a put; and put on a put. The option has a first period and strike level, and a second period and strike level.
- Forward Start Option is an option, which will start at an agreed date in the future. At this time the strike level will be set here at the money. The option is then a standard one.
- Lookback Option is an option where pay off at expiry uses the historical price evolution of the underlying stock over the life of the option to create the most favourable payout (i.e. the payout depends on the maximum or minimum stock price during the life of the option).
- Shout Option is an otherwise Standard European option in which on an agreed set of dates or on the basis of a limited number of opportunities (including expiry) the option holder can lock in the then current stock price to create a positive or more positive payout at expiry. It is in effect a limited form of look back option and will lie in value between a standard European Option and a look back option depending upon the number of "shouts" involved

[9]

ii) **Key risks**

- Enforceability: Whether derivatives can be deemed legal transactions, and not gambling

- Capacity: Whether entity entering into the transaction, and the person actually arranging the deal, have the power and authority so to do any other broad limitation on the instrument used.
- Documentation: The status of written documentation is important some jurisdictions allow verbal confirmation and there can be uncertainty when verbal and written confirmations disagree.
- Credit risk mitigation: A collateral agreement would help reduce credit risk. But the trade valuations will need to be agreed with the counterparty not easy if they are complex.

[5]

[14 Marks]**Soultion 2:**

- i) Considering the variable $y(t, T) = \ln P(t, T)$ using Ito's lemma we get

$$dy = d(\ln P) = \left[r - \frac{1}{2} \frac{\sigma^2}{P^2} \right] dt + \frac{\sigma}{P} dz = \left[r - \frac{1}{2} v^2 \right] dt + v dz$$

The time t forward rate spanning period T_1 and T_2 is given by

$$f(t, T_1, T_2) = - \frac{y(t, T_2) - y(t, T_1)}{T_2 - T_1}$$

Substituting first equation gives the expression for the forward rate. And letting T_2 to T_1 we get the instantaneous forward rate.

This again leads to $(..)[dt] + (..)[dz]$ form which is depended on v and its derivatives hence there is an explicit link between the volatility of zcb with forward rate drift.

[8]

- ii) Benefit of 2 factor HJM model:

HJM is a generalised framework for modelling the yield curve evolution, enabling a wide range of valuations from one model, i.e. efficient and consistent.

Two factors enable the model to describe greater variations in the evolution of the yield curve, since forward rates can be imperfectly correlated.

Two factors also enable valuation of options which depend not just on the market level but also on the yield curve slope,

Choice of factors:

The method of principal components can be used to identify the main drivers of the yield curve which generally are the direction and the slope.

Implementation and calibration:

It is difficult to implement the HJM model based on market prices. Prices of caps or swaptions to drive model parameters, though these are values in the market using the Black model, which is a lognormal distribution and specifying a log-normal volatility for forward rates in HJM severely restricts the forms of the HJM model that can be used.

Choice of correlation function is difficult - hard to be precise about it since so few reliable option values to base it on. Swaptions and caplets imply certain correlation effects but, considering the calibration difficulties, are not likely to suggest a good correlation function.

[6]

[14 Marks]

Solution 3:

	Annual Coupon	Term to Maturity	Clean Price
Creta LTD	7%	1	99
Statum LTD	5%	2	98

Interest rate	5.00%
Nominal	100

i)

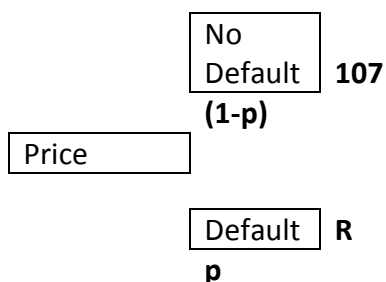
Creta LTD	Payout	Discounted
1	107	101.9048

Statum LTD	Total	100
1	5	4.761905
2	105	95.2381

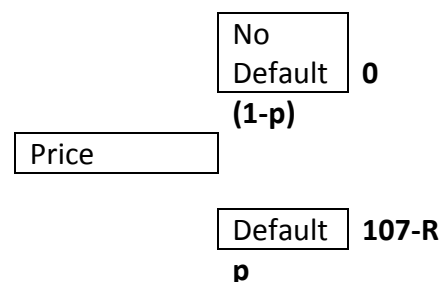
[2]

ii)

Creta LTD



CDS



R is recovery rate per bond

Consider Buy bond and Sell
CDS

	No Default	107
	(1-p)	
Price		
	Default	107-R+R
	p	

The above portfolio is risk neutral hence

Hence price of the CDS is price of equivalent risk free bond – price of the corporate bonds

CDS 2.904761905

[3]

- iii) For 2-year bond similar to the 1-year bond price of equivalent risk free bond – price of the corporate bond.

[2]

- iv) The difference in pricing could be due to the riskiness of the bond, assigned credit rating, past default experience, sector type, debt to equity ratio of the company issuing the bond, liquidity of the bond in the market.

[3]

[10 Marks]

Solution 4:

- i) The continuously compounded 3.75 year zero coupon rate can be calculated by interpolating the rates between 3 and 4. The solution gives linear interpolation though any other approach is acceptable

Rate at point	3.75
Rate	7.875
Bond price	0.753

[2]

- ii) Fixed leg coupon of a 4-year par value annual interest rate swap can be calculated by making the swap cost neutral at t-0.

2	4 year par swap			factor
	Rates	Rate	Discount	Name
	1	7.25	0.932401	d1
	2	7.35	0.867753	d2
	3	7.65	0.8016	d3
	4	7.95	0.736393	d4
	5	8.2	0.674316	d5
	6	8.35	0.618054	d6
	7	8.5	0.564926	d7
	Given par coupon			
	coupon* ($\sum$ Discount) =1-Discount (t=4)			
	Coupon	7.8968%		

[2]

- iii) The fixed leg coupon of a forward-starting 4-year par value annual to annual interest rate swap commencing in 3- years' can be calculated by making the swap cost neutral at year 3.

Starting in 2 year 5 year swap			
Let the coupon be Cf			
Swap need to be cost neutral at t=0 hence			
$Cf(d_4+d_5+d_6+d_7)=f_4.d_4+f_5.d_5+f_6.d_6+f_7.d_7$			
where f_n is the forward rate			
$f_n=(d_{n-1}/d_n)-1$			
Coupon	0.09124994		

[3]

[7 Marks]

Solution 5:

- i) If a call futures option is exercised, the holder receives a long position in the underlying futures contract plus an amount of cash equal to the current futures price minus the strike price.

If a put futures option is exercised, the holder gets a short position in the underlying futures contract plus an amount of cash equal to the difference between the strike price and the current futures price.

[2]

- ii) The price of a stock falls when a dividend is paid; the fall being roughly equal to the amount of the dividend. Payment of a continuous dividend at a rate q causes the growth rate of the stock to be less than it would otherwise be. Suppose that with a continuous dividend yield of q , a stock price grows from S at time t to S_T at time T . Then if there were no dividends, it would grow from S at time t to $S_t e^{q(T-t)}$. Thus the standard BS formula can be altered by replacing S with $S_t e^{q(T-t)}$

$$C = S e^{-q(T-t)} N(d_1) - X e^{-r(T-t)} N(d_2)$$

$$P = X e^{-r(T-t)} N(-d_2) - S e^{-q(T-t)} N(-d_1)$$

[3]

[5 Marks]

Solution 6:

- i) The basic formula to be used is the Black Scholes formula – in particular for the put option, it is given as

$$p = X e^{-rT} N(-d_2) - S N(-d_1)$$

where

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \text{ and } d_2 = d_1 - \sigma\sqrt{T}$$

Also,

- S is spot price, i.e. price of the basket today
- X is strike / exercise price
- r is one-year continuously compounded risk-free rate
- T is term of option = 1 year
- σ is volatility – it may be measured through the implied volatility approach

Allow for dividends

The Black-Scholes formula above assumes that the underlying asset produces no income. In reality the equities will produce dividends, so an adjustment to the formula is necessary to allow for this.

Possible methodologies are:

- To assume dividends are paid continuously at rate q and to replace S with $S e^{-qT}$ to get

$$p = X e^{-rT} N(-d_2) - S e^{-qT} N(-d_1)$$

and

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r - q + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \text{ and } d_2 = d_1 - \sigma\sqrt{T}$$

- But better (given that there are only two equities) would be to allow for dividends explicitly by replacing S with $S - \sum_t \text{div}_t e^{-rt}$ the sum of the present value of all dividends; i.e.

$$\text{Replace } S \text{ with } S - \sum_t \text{div}_t e^{-rt}$$

Risk-free rate: r would be derived from a yield curve that reproduces swap rates.

Implied volatility: Implied volatilities of the two equities (σ_1 , σ_2) can be derived separately by looking at prices of one-year (put or call) options on the two equities and back solving.

Historical volatility for each of the equities could be used to help set σ_1 and σ_2 .

Looking at trends in historic volatility and considering future events can be used to convert the historic measure into a forward looking parameter for pricing.

The bank needs a methodology for combining σ_1 and σ_2 ; the volatilities for each of the assets into a basket volatility σ .

A common approach would be to use the correlation between the equities ρ . Given individual volatilities σ_1 and σ_2 and spot prices S_1 and S_2 (excluding dividends, as stated earlier) then the basket volatility σ can be determined by solving for the value σ which equates the combined lognormal variances of the underlying equities (allowing for their correlation) with an assumed lognormal variance of the basket.

Or the bank might decide to use a simpler approximation such as

$$(S_1 + S_2)\sigma^2 = S_1\sigma_1^2 + S_2\sigma_2^2 + 2\rho S_1S_2\sigma_1\sigma_2$$

Correlation: The bank also needs to set the correlation parameter ρ .

Correlation will vary over time and over varying market conditions.

If there are no traded options on the basket, as is likely, then a market implied correlation will not exist. This leaves historic experience as the main driver of a correlation assumption.

However, because it is unhedgeable, and because price changes of the two equities in the basket will tend to increase in size and codependency in market crises where the put option is likely to be heavily in the money, the bank will probably include a safety margin.

The use of a simplified formula for basket correlation might be another reason to include an extra margin within the value chosen for ρ .

Margins: Finally, the bank will, in all likelihood, include margins in the price for:

- Profit
- Expenses
- And, in all likelihood, Capital costs or hedging costs.

[9]

ii)

a) The Greeks that the bank will be most keen to hedge are

$$\begin{aligned} \text{Two Deltas: } & \frac{\partial p}{\partial S_1} \text{ and } \frac{\partial p}{\partial S_2} \\ \text{Two Gammas: } & \frac{\partial^2 p}{\partial S_1^2} \text{ and } \frac{\partial^2 p}{\partial S_2^2} \\ \text{Two Vegas: } & \frac{\partial p}{\partial \sigma_1} \text{ and } \frac{\partial p}{\partial \sigma_2} \\ \text{Rho: } & \frac{\partial p}{\partial r} \\ \text{and Cross Gamma: } & \frac{\partial^2 p}{\partial S_1 \partial S_2} \end{aligned}$$

b) The possible hedging instruments include:

- Futures / forwards on the individual equities in the basket
- Put options on the individual equities in the basket
- Call options on the individual equities in the basket
- Options will ideally be one-year options with similar strikes (expressed as % of spot price) to the option in question
- One year zero-coupon bonds
- Swaps
- Volatility based derivatives – in particular volatility index futures
- Underlying assets

c) The relevant signs are:

	Futures	Put Option (Reverse signs for short puts)	Call Option (Reverse signs for short calls)	Zero Coupon Bonds or Receiver Swaps
Delta	Positive	Negative	Positive	No effect
Gamma	No effect	Positive	Positive	No effect
Vega	No Effect	Positive	Positive	No effect
Rho	Positive	Negative	Positive	Negative

[12]

- iii) Changes to the implied correlation between the two equities will be difficult to hedge. Even if another bank is prepared to create a hedging instrument, it will require compensation for writing an unhedgeable risk of its own.

The first step (because options will change the rho and delta of the hedged position) will be to hedge out the gammas and vegas using options.

The bank will need to buy four options to do this. Because a single option is unlikely to hedge out both gamma and vega at the same time, the bank will need two options on each of the shares in the basket.

Next the deltas need to be hedged (as futures will have an impact on the rho of the hedged portfolio).

The bank will calculate the deltas of the shorted OTC put and of the long positions in options (that are hedging out the gammas and vegas). It will calculate how many futures to enter into (which could be short or long positions) to cancel out these deltas.

Finally, the bank will hedge rho.

Because it has a large derivative position already on its books, it probably already hedges against yield curve movements. So rather than hedge this deal separately, the rho exposure within the deal (and the hedges described) will be incorporated into its overall yield curve hedging processes.

One possible way this is likely to work is that the bank will already have systems to:

- Convert market swap rates to a zero coupon yield curve
- Value its balance sheet based on the yield curve.

With this system in place, the bank can test the sensitivity of its balance sheet to changes in individual swap rates and use this to determine any extra swaps it needs to enter into to hedge out its exposure to movements in the yield curve.

[5]

[26 Marks]

Solution 7:

i)

- a) Reasons for choosing a full yield curve model:

The rationale for a term structure yield curve model is to be able to price simultaneously options spread across the entire range of maturities in a yield curve.

Exotic interest rate swaps and options include spread options, Bermudan swaptions and path dependent options. These depend not just on the evolution of forward rates along the curve, but on the correlation between changes.

b) Desirable features:

- Model is arbitrage free, so produces prices consistent with the market
- Flexible enough to cope properly with the required range of derivative contracts
- Easy to specify and calculate
- The current swap (or bond) curve should be reproduced by the model
- Easy to calibrate
- Enough degrees of freedom (parameters) to make the model flexible to cope with any yield curve shape, but not overly flexible so there is instability between parameters from one day to the next
- Volatility of rates of different maturity should be different, with shorter rates usually seeing higher volatility
- Imperfect correlation between forward rates
- Negative interest rates should not normally be allowed
- Mean reversion of interest rates
- Reasonable dispersion of rates over time

c) No – arbitrage models:

No-arbitrage, or “arbitrage free”, models are a class of models that allow recovery of market prices of one set of securities given prices of another set. This creates a world of “relative” prices.

In a non-“arbitrage free” model, securities could be priced using the model and then traded at a different price in the real world, leading to persistent profit – something not tenable in the financial markets.

No-arbitrage is very important in yield curve models, since most complex structures are limiting cases of simpler structures (such as swaps, caps, floors) and ideally the model should recover the prices of the latter exactly.

Also, hedging is done using the simpler structures, so the accounting process will not be distorted by imaginary gains and losses.

[9]

- ii) A trinomial tree is a discrete-time representation of the continuous rate process, with the stochastic process defined by branches which can have three states: up, down and mid-way between.

If the probabilities of going up, down and middle are p_u , p_d and p_m respectively, then $(p_u + p_d + p_m) = 1$ and all values are strictly positive.

The time horizon is split into constant intervals of time Δt , so the rates are compounded Δt period rates.

Further, accuracy is improved with smaller time-steps, up to a point.

The tree for the one-factor model is built in two stages.

Stage 1:

First, set $b(t)$ to zero. Then,

$$dr^* = -ar^*dt + \sigma dz$$

which gives a rate process r^* that is initially zero, and whose evolution is governed by a constant mean reversion towards zero.

The value of σ is determined from the caplet volatilities.

There are three unknowns: p_u , p_d and Δr^* (since $p_m = 1 - p_u - p_d$).

It is convenient to set $\Delta r^* = \sigma\sqrt{3\Delta t}$, since this has been found to give the best numerical efficiency.

The other two equations come from the expectation and variance of Δr^* over the interval Δt , in which the tree must match the process.

There are three different forms of the mean and variance equations for the tree: when all the branches point upwards or horizontal, when all point downwards or horizontal, and when they straddle the line of unchanged rates.

There will be different probabilities for each situation.

Since $E(\Delta r^*)^2$ contains the term $\sigma^2\Delta t$, the equations account for the volatility component.

Stage 2:

Now add back in the term structure, to move back from r^* to r .

Let $\alpha(t) = r(t) - r^*(t)$. Then:

$$d\alpha = a[b(t) - \alpha(t)]dt \quad (1)$$

which is a deterministic mean-reverting process (i.e. varies over time but not stochastically), which can easily be calibrated to the term structure.

The best method of solving the differential equation in (1) is by forward induction along the Δt scale using Green's functions. A Green's function $Q_{i,j}$ is the present value of 1 unit of cash payable at node (i, j) on the tree.

Now, $Q_{0,0} = 1$ and $\alpha_0 (= r)$ is set in terms of the price of a zero-coupon bond maturing at Δt .

Then $Q_{1,1}$, $Q_{1,0}$ and $Q_{1,-1}$ are set in terms of the tree probabilities p_u , p_d and p_m , $Q_{0,0}$ and α_0 .

One can then easily find the value of α_1 based on the price of a zero-coupon bond maturing at $2\Delta t$ and these Q 's.

One can then move up the time horizon.

[10]

iii) Does HW fulfil the desirable features?

The model is a suitable no-arbitrage full yield curve model as

- It reproduces vanilla bond and swap prices exactly and prices options on these
- The model incorporates mean reversion and time-dependent parameters, so is flexible, behaves well and has little chance of producing negative rates
- It is relatively easy to specify mathematically, though some care is required.

However, a few issues that need to be mentioned are

- It only has one driving factor, so all forward rate moves have to be completely correlated and hence it cannot re-create complex yield curve changes
- The underlying distribution of the interest rate is normal, not log-normal, so it can lead to negative rates [theoretically, as in practice the probabilities of negative rates are negligible]
- The model may be hard to calibrate to cap prices which are priced in the market with the Black's log-normal model.

Also,

- Differences arise as interest rates move away from the current level. These effects, if not disentangled, will lead HW to have an incorrect balance between its volatility and mean-reversion parameters.
- Using constant values for α and σ limits the shapes of yield curve that can be modelled. Time-dependent mean reversion and volatility parameters can fit any current yield curve and forward volatility "hump" shape although they can imply an implausible evolution of the term structure of volatility.

[5]

[24 Marks]
